

ART Maritime High Power Ultra High Frequency (UHF) Emitter System SOW

1.0 Background

The Aircraft Signature and Avionics Measurements (ASAM) Electronic Warfare (EW) group of the Atlantic Ranges and Targets (ART) department operates numerous pulsed radio frequency (RF) mobile emitter systems that can be deployed on land or installed on boats. The mobile emitter systems support research, development, test and evaluation (RDT&E) requirements of military fixed-wing and rotary-wing aircraft as well as Unmanned Aircraft Systems (UAS). The current mobile emitter systems fit into one of two general categories, which are high power or low power. The high power systems meet most customer demands in regard to effective radiated power (ERP) but have limited utility due to low duty cycles and insufficient pulsed waveform capabilities. The low power systems are able to achieve much higher duty cycles and generate complex pulsed waveforms but are limited due to the lower ERP they can produce. To support aircraft RDT&E EW requirements into the future, a maritime high power UHF system that is also capable of handling high duty cycles and producing complex waveforms has been identified as essential.

2.0 Scope

The purpose of this document is to define the key requirements for a new maritime high power UHF emitter system. The requirements are traceable to RDT&E EW user needs.

3.0 System Requirements

The emitter system will be permanently installed onboard a 125 foot boat and will be capable of operating in the open ocean. The design must include an air conditioned enclosure to house all of the equipment in order to provide protection from the elements and cooling while in operation. Additionally, the system is required to have its own generator to supply all electrical power needs during a mission. The antenna design must include two phased array panels, one mounted on each side of the boat facing either port or starboard. The expectation is that the system will be delivered with manuals and technical drawings and training will be supplied after government acceptance. Spares for any high failure items such as antenna components and amplifier modules should also be part of submissions. Additional RF performance requirements are located in Table 1.

Table 1. High Power UHF Emitter Requirements

Name	Requirement	Comment
Frequency Range	400 – 500 MHz	
Pulse Repetition Frequency (PRF)	Up to 25 kHz	
Pulse Width (PW) Range	2 - 2000 us	
Modulation on Pulse	Both frequency & phase modulation	
Duty Cycle	25%	
ERP	104 dBm	
Antenna Scan	Sector / Uni-directional 1 – 30 sec scan period	Each array at a minimum capable of sector and uni-directional scans over an azimuth range of 90 deg
Antenna Polarization	Horizontal	
Antenna Horizontal Beamwidth	Greater Than 6 deg	
Signal Source	Programmable across full range of RF, PRF/PRI, PW & pulse modulation types	

4.0 Deliverables

One high power UHF emitter system meeting or exceeding the requirements outlined in section 3.0 with associated manuals, drawings and spares.

5.0 Delivery

All items shall be received by the government within one year from contract award.